

PRODUCT STRATEGY FOR THE COMPUTATIONAL INJURY ANALYSIS GROUP

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ABSTRACT. DRAFT - WORK-IN-PROGRESS - NOT FOR RELEASE

1. PURPOSE

The purpose of this brief is to articulate a product strategy for the Computational Injury Analysis Group within Terminal Ballistic Department (Department 5421) at Sandia National Laboratories.

The intended audience for this brief is Douglas Dederman and the author.

This brief is in response to Wallace Bow (see email 2016-10-31 1433 MST), which stated that the Center (Integrated Military Systems, Organization 5400) solicits this brief from its organizational reports so that the Center can create a list, called STIPL, Science and Technology Integrated Priority List, which then allows the Center “to define and prioritize our most pressing technological issues to guarantee that our innovation investments are being properly allocated.” The end goal of effort and funding allocation is “to ensure that our products remain superior.”

2. BACKGROUND

To effectively articulate a product strategy for the Computational Injury Analysis Group, some background is helpful.

2.1. **Who.** The group is led by Douglas A. Dederman (Manager of Department 5421). Paul A. Taylor is the senior member. Candice F. Cooper, Shivonne Haniff, and Chad B. Hovey are members.

2.2. **What.** The group has product offerings in modeling and simulation, and scientific publications. The group’s model is a high-fidelity head, neck, and torso human body model.

2.3. **When.** There have been two main Laboratory Directed Research and Development (LDRD) grants, the first in FY2007 for one year and the second in FY2013 for three years.

2.3.1. *Traumatic Brain Injury.*

- FY2007 marked an investment of \$99,000 sponsored by Defense Systems and Assessments (Division 5000), for Paul A. Taylor, Principal Investigator, John B. Aidun, Product Manager, to investigate a connection between blast and traumatic brain injury (07-1614 Proposal Summary, LDRD Project 113675 “Traumatic Brain Injury”). The \$99,000 funding amount was recorded as 0.30 Full Time Equivalent (FTE) in Center 1400, with zero funding requests for out years FY2008 and FY2009.

2.3.2. *Wound Ballistics Modeling for Blast Loading, Blunt Force Impact, and Projectile Penetration.*

- FY2013 marked a funding request of \$380,000 for labor (Taylor at 0.5 FTE, Ludwigsen at 0.1 FTE, Brundage at 0.1 FTE) and \$35,000 for acquisition of medical image data and hiring part-time students to segment the medical images, total \$415,000.
- FY2014 requested \$392,000 for labor at the same FTE allocation as in FY2013 (\$12,000 or 31.6 percent YoY increase) and \$4,000 for miscellaneous supplies, total \$396,000.
- FY2015 requested \$396,000 for labor at the same FTE allocation as in FY2014 (\$4,000 or 1.02 percent YoY increase), \$4,000 for miscellaneous supplies, and \$4,000 for travel to present at scientific conferences, total \$404,000.
- Total for second LDRD investment: \$1,215,000.
- Total for both LDRD investments: \$1,314,000.

See Taylor IDAPR LDRD Quad New_Initial.v7.pptx and LDRD Project 165554.

2.4. Other investments. ONR funding (Bentley, amount, time) to be determined. Other funding to be determined.

3. CONTEXT

In this section, we answer the questions, “What is the group’s strategy and how does it integrate with the larger mission and vision of Sandia National Laboratories.”

3.1. Sandia National Laboratories. Sandia’s mission states, “The synergy and interdependence between our nuclear deterrence mission and broader security missions forge a robust capability base and empower us to solve complex national security problems.”¹

Sandia’s vision states, “On behalf of our nation, we anticipate and solve the most challenging problems that threaten security in the 21st century.”

¹See <http://strategicplan.sandia.gov/vision-mission-values.html>.

3.2. Defense Systems and Analysis. Within Sandia, the Defense Systems and Assessments (DSA, Division 5000) states its mission as “We shape the future of national security by delivering tools, technologies, and pathfinder solutions that exploit the vulnerabilities of U.S. adversaries. By leveraging the Lab’s research and development in advanced science, technology, and systems engineering we create and advance capabilities that are synergistic with our nuclear weapons mission.”²

DSA’s vision states, “We transform research into game changing solutions that provide the warfighter and intelligence community with unmatched superiority.”

3.3. Integrated Military Systems. Within DSA, Integrated Military Systems (IMS, Organization 5400) states its mission as “Integrated Military Systems develops and demonstrates game-changing systems and technologies to solve our warfighters’ most challenging, urgent, and high-risk problems—to defend our nation and defeat threats to national security.”³

IMS states its vision as “Render exceptional service in the national interest and seize opportunities that protect our warfighter and promote national security.”

3.4. Strike and Aerospace Systems. Within IMS, the Strike and Aerospace Systems (Organization 5420) states its mission as “Support the DoD and other Federal agencies at Sandia’s NWSBU Advanced Systems and Technology Program with technologies focused on high-speed reentry, precision, navigation, earth-penetrating weapons, conventional munitions, and guided flight systems.”⁴

3.5. Terminal Ballistics Technology. Terminal Ballistics Technology (TBT, Organization 5421) states its mission as “Design, develop, and demonstrate terminal ballistics enabling technologies that enhance survivability and functionality of lethal and nonlethal projectiles/probes for characterization, defeat or protection of military and nonmilitary targets.”⁵

TBT has a programmatic relationship with Military Systems and Technologies (MST, Organization 5440). MST states “The people of 5440 apply science-based systems engineering, integrated technologies, and advanced analytics to enhance military operations and mission effectiveness.”⁶

²See <https://sharepoint.sandia.gov/sites/dsa-div-5000/Pages/default.aspx>.

³See https://sharepoint.sandia.gov/sites/IntegratedMilitarySystems/About_Us.aspx.

⁴See http://info.sandia.gov/imsdc/dept15420/5420_home.htm.

⁵See <http://info.sandia.gov/imsdc/dept5431/index.htm>.

⁶See http://info.sandia.gov/imsdc/dept5440/5440_home.htm.

4. MISSION AND VISION

Many of the themes embodied in the above missions entail service to national security by solving the problems of the U.S. warfighter. Within that scope, we state our mission and vision.

4.1. Mission. We analyze injury from blast and ballistics to help the nation protect the U.S. warfighter.

4.2. Vision. To significantly reduce the risk of death or serious injury from blast and ballistics by 2027.

Why computational? Because the core competencies of our group is within modeling and simulation. By this measure, we, by design, exclude experimental work from efforts we undertake. Our interaction with experimental groups, therefore, is collaborative.

Why injury? Because injury, or specifically, the reduction or elimination of injury, is fundamental to the nation's ability to marshal national security resources that require a human in-the-loop. We emphasize the word "injury" because the analysis goes beyond demonstration of engineering response (e.g., stress, strain) of human tissue and human personal protection (e.g., bullet proof vest) to war environments. The analysis connects engineering response to human threshold of injury to provide evidence-based prediction on injury severity: minor, moderate, serious, severe (survivability likely) critical (survivability unknown), and maximum (not survivable). See the Abbreviated Injury Scale, Association for the Advancement of Automotive Medicine.

Why analysis? Because our main product is the advancement of science to understand injury mechanisms. **Why blast and ballistics?** Because the exposure of the warfighter to bombs and high velocity munitions continue to be a specific and relevant risk to the U.S. warfighter. We emphasize blast and ballistics and distinct from general impact because the characteristic forces and velocities of the former very high, the time scales over which the events occur are extremely short, creating an environment that is extremely hostile to human viability.

Why by 2027? Because we believe pursuit of significant reduction to the risk of death or serious injury to the U.S. warfighter will remain an enduring research endeavor over the next decade, worthy of considerable investments of time and resources. It is interesting to conceive of autonomous systems (e.g., robots) taking the place of a human soldier on the war field. In this scenario, our mission may evolve, but our vision would remain unchanged. Our mission, therefore, might shift to protection of the human surrogate from malfunction or obliteration secondary to exposure to blast or ballistics. In this way, we have attempted to be forward-thinking in our vision, while calibrating a mission to important, present-day needs.

5. STRATEGY

In a global context, we direct our efforts to protect people who promote peace. The specific implementation of this ideal is through the U.S. warfighter, called upon by service to country to fight tyranny and the abuse of power. "Life, liberty and the pursuit of all who threaten it," the U.S. Navy's current slogan, adroitly encapsulates a warfighter's mission, and to that end, it is incumbent on those who benefit from a warfighter's risk taking that we help to minimize exposure to that risk. Specifically, as long as blast and ballistics remain an inherent element of armed conflict, we should equip the warfighter with the best available protection from the same.

Fundamental to the best available protection paradigm is the understanding of how blast and ballistics imperil the warfighter's survivability by exposing the warfighter to traumatic brain injuries that are either focal (penetrating through the warfighter's helmet or face, impact to a side of the helmet or face) or diffuse (causing localized shear damage to the neural microstructure of the brain).

Our strategy is build from our extraordinary success since our first funding in FY2007. However, before we explicate the path forward, it is important to lay out, in an abbreviated form, the foundation from which we build and articulate how our group is unique in the world and thus positioned to offer the nation a significant value proposition.

What is our product, as we have built it to date? We have developed and validated human tissue constitutive models for use in Sandia's Sierra and CTH computational code bases. We have developed and validated a human anatomical model of the head, neck, and torso, having 1 millimeter resolution, based on the Visible Human Project. We have published peer-reviewed journal articles that have postulated that critical thresholds of deviatoric strain energy in the human brain secondary to blast correlate with traumatic brain injury (TBI) and clinical functional magnetic resonance imaging (fMRI) data that suggest the brain has a compensatory mechanism when injured.

Why have we focused on the head, neck, and torso? Because our strategy focuses our limited resources on serious-to-fatal injuries. While fatal injuries can still occur secondary to injury to the upper or lower extremities, the prevalence in these body regions is relatively low compared to injuries to the head, neck, and torso. Therefore, our strategy has guided us to enumerate autonomous systems (e.g., robots) taking the place of a human soldier on the war field. In this scenario, our mission may evolve, but our vision would remain unchanged. Our mission, therefore, might shift to protection of the human surrogate from malfunction or obliteration secondary to exposure to blast or ballistics. In this way, we have attempted to be forward-thinking in our vision, while calibrating a mission to important, present-day needs.

What the current efforts that continue to improve our product? Our team is actively working to validate the torso regions of the model with known viscous injury criteria,

6. COMPETITIVE LANDSCAPE

How are we uniquely positioned?

7. PURPOSE